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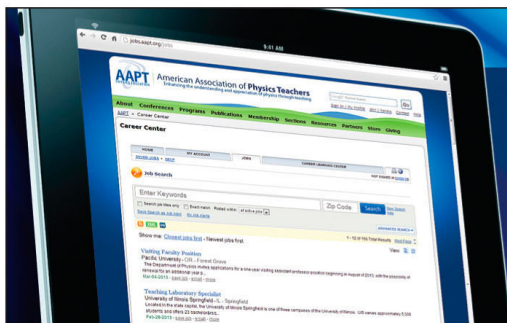
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A toy model for the yield of a tamped fission bomb

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A simple expression is developed for estimating the yield of a tamped fission bomb, that is, a basic nuclear weapon comprising a fissile core jacketed by a surrounding neutron-reflecting tamper. This expression is based on modeling the nuclear chain reaction as a geometric progression in combination with a previously published expression for the threshold-criticality condition for such a core. The derivation is especially straightforward, as it requires no knowledge of diffusion theory and should be accessible to students of both physics and policy. The calculation can be set up as a single page spreadsheet. Application to the *Little Boy* and *Fat Man* bombs of World War II gives results in reasonable accord with published yield estimates for these weapons. © 2018 American Association of Physics Teachers.

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I. INTRODUCTION

This paper continues a series of papers previously published in this journal concerning the physics of Manhattan Project-era fission bombs (see Refs. 1–3 and references within; Ref. 4 is a recent Resource Letter on the Manhattan Project and related issues). The present work describes the development of a simple yet reasonably accurate expression for estimating the yield and efficiency of a tamped fission bomb that may or may not be compressed (i.e., imploded) when the chain reaction is initiated. As described in previous papers, a tamper is a metal jacket surrounding the fissile core of the bomb, the purpose of which is to reflect escaping neutrons back into the core to give them further chances at causing fissions and thereby reduce the necessary critical mass of fissile material while enhancing the efficiency of the explosion. The *Fat Man* bomb detonated at the *Trinity* test of July 16, 1945 and later deployed at Nagasaki was a plutonium-based tamped implosion device, whereas the uranium-based *Little Boy* “gun bomb” deployed at Hiroshima employed a tamper but not implosion. The advantage of implosion compression is that because the critical radius of the (assumed spherical) core is inversely proportional to its density, an implosion weapon can begin with a subcritical core that is made critical by crushing it to a higher density, and this reduces the amount of valuable fissile material required to achieve an explosion of a given yield.

While yield is probably the most important metric of a nuclear explosion, it is a notoriously difficult quantity to estimate accurately. The evolution of rate of fissions, number of neutrons, pressure, energy density, and core expansion speed all vary exponentially with time. Analytical approximations based on diffusion theory can be developed, but detailed accuracy requires time-driven numerical simulations run on high-speed computers, which were certainly not available during the Manhattan Project. Given the importance of estimating yield, having available a simple model that is easy to use while giving reasonable estimates should be of value to students of both physics and non-proliferation policy. The purpose of this paper is to develop such a model. In addition to its ease of use, the model developed here has the further advantage that users need have no knowledge of neutron-transport theory in order to grasp the essence of the derivation. The method does utilize a key result from diffusion analysis, but this can be presented and explained without having to delve into the underlying derivation.

The result required from diffusion theory involves the fact that as the chain reaction proceeds within a bomb core, the core rapidly heats up, vaporizes, and becomes a high-pressure gas (see Sec. 3.1.4 of Ref. 5). The core will inevitably expand, and its density must consequently drop. The condition of criticality—the physical conditions necessary for the chain reaction to keep propagating—depends on the density, which means that the core will reach a radius at which criticality shuts down. After this time, essentially no more energy will be released. In technical parlance, this condition is known as “second criticality”; this terminology derives from defining the moment when conditions within the core can initially support criticality as being that of “first criticality.” For the present purpose, the important quantity is the radial distance $\Delta R = R_{\text{shut}} - R_{\text{init}}$ through which the core expands in going from its initial radius R_{init} when the chain reaction is initiated to its second-criticality radius R_{shut} . For a core and tamper configuration of given materials and masses, R_{shut} can be determined from an expression arising from diffusion analysis (as discussed below).

II. (CORE + TAMPER) MODEL

Consider a spherical core of fissile material of mass M_{core} snugly surrounded by a spherical tamper of mass M_{tamp} as sketched in Fig. 1. Let the normal densities of the core and tamper materials be ρ_{core} and ρ_{tamp} , that is, these are their uncompressed or unexpanded “normal laboratory condition” densities. To account for an implosion weapon, define the compression ratio C as the ratio of the densities of the core and tamper at any time to their normal densities, with C_{init} being the value of this ratio at the moment the chain reaction is triggered. The assumption that both the core and tamper are compressed by the same ratio is a simplifying approximation for pedagogical purposes; a refinement might be to relate compression ratios via bulk moduli. For a non-imploded weapon, $C_{\text{init}} = 1$; implosion corresponds to $C_{\text{init}} > 1$. The initial radii of the core and tamper will then be

$$R_{\text{core}} = \left(\frac{3 M_{\text{core}}}{4\pi C_{\text{init}} \rho_{\text{core}}} \right)^{1/3} \quad (1)$$

and

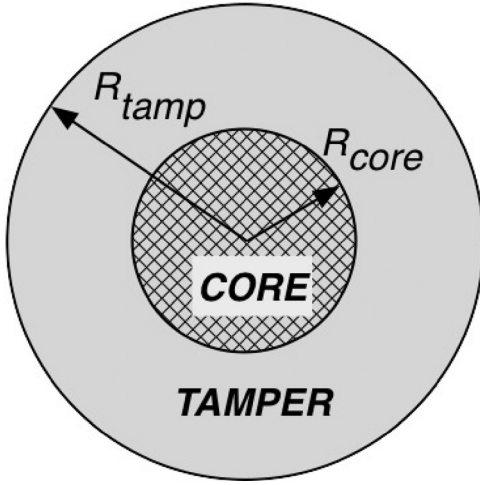


Fig. 1. Schematic illustration of a fissile core and surrounding tamper.

$$R_{\text{tamp}} = \left[\frac{3}{4\pi C_{\text{init}}} \left(\frac{M_{\text{tamp}}}{\rho_{\text{tamp}}} + \frac{M_{\text{core}}}{\rho_{\text{core}}} \right) \right]^{1/3}. \quad (2)$$

As the core and tamper expand during the explosion, C will drop, but I assume that both components always have the same compression ratio at all times. Let $R_{\text{core}}^{\text{shut}}$ and $R_{\text{tamp}}^{\text{shut}}$ be the core and tamper radii at second criticality, at which time the compression ratio is C_{final} . As shown in Ref. 2, the condition of second criticality is given by the constraint that

$$\left[1 + \frac{2R_{\text{core}}^{\text{shut}} \lambda_{\text{tamp}}^{\text{trans}}}{3(R_{\text{tamp}}^{\text{shut}})^2} - \frac{R_{\text{core}}^{\text{shut}}}{R_{\text{tamp}}^{\text{shut}}} \right] \left[\left(\frac{R_{\text{core}}^{\text{shut}}}{d} \right) \cot \left(\frac{R_{\text{core}}^{\text{shut}}}{d} \right) - 1 \right] + \frac{\lambda_{\text{tamp}}^{\text{trans}}}{\lambda_{\text{core}}^{\text{trans}}} = 0, \quad (3)$$

where $R_{\text{core}}^{\text{shut}}$ and $R_{\text{tamp}}^{\text{shut}}$ are calculated with Eqs. (1) and (2) but with C_{final} replacing C_{init} . Once the value of C_{final} that renders Eq. (3) satisfied has been determined (this can only be done numerically), the core radius at second criticality is determined from Eq. (1), and the radial expansion distance ΔR follows.

All of the other quantities in Eq. (3) have dimensions of distance. The quantity d involves only core parameters and is given by

$$d = \sqrt{\frac{\lambda_{\text{core}}^{\text{fiss}} \lambda_{\text{core}}^{\text{trans}}}{3(\nu - 1)}}. \quad (4)$$

The quantity ν here is the (dimensionless) number of neutrons liberated per fission, and $\lambda_{\text{core}}^{\text{fiss}}$ and $\lambda_{\text{core}}^{\text{trans}}$ are the mean free paths for fission and neutron transport, respectively,

$$\lambda_{\text{core}}^{\text{fiss}} = \frac{1}{C_{\text{final}} \sigma_{\text{fiss}} n_{\text{core}}} \quad (5)$$

and

$$\lambda_{\text{core}}^{\text{trans}} = \frac{1}{C_{\text{final}} (\sigma_{\text{fiss}} + \sigma_{\text{elas}}) n_{\text{core}}}, \quad (6)$$

where σ_{fiss} and σ_{elas} are the reaction cross-sections of the core material for fission and elastic scattering of neutrons, respectively.

The quantity n_{core} in Eqs. (5) and (6) is the nuclear number density of the core material corresponding to its normal mass density. If A is the atomic weight of the material in g mol^{-1} and N_A is Avogadro's number, then n in nuclei cm^{-3} is given by $\rho N_A / A$. Finally, $\lambda_{\text{tamp}}^{\text{trans}}$ in Eq. (3) is analogous to Eq. (6) but for the tamper material, which is assumed to scatter neutrons only elastically

$$\lambda_{\text{tamp}}^{\text{trans}} = \frac{1}{C_{\text{final}} \sigma_{\text{tamp}}^{\text{elas}} n_{\text{tamp}}}. \quad (7)$$

A quantity that will prove important in the yield analysis of the next section is the time τ required for a neutron born in a fission to subsequently strike and fission another nucleus. If the average speed of fission-born neutrons is v_{neut} , this time is given by

$$\tau = \lambda_{\text{core}}^{\text{fiss}} / v_{\text{neut}}. \quad (8)$$

Clearly, τ will be a function of the continuously evolving compression ratio C ; as C falls, τ must increase. How this is dealt with is addressed in Sec. III.

III. YIELD AND EFFICIENCY MODEL

Chain reactions are essentially geometric-growth phenomena, and this behavior can be used to develop an estimate of the energy liberated by an exploding bomb core up until the time it reaches second criticality. Suppose that N_0 “initiator” neutrons are available to start the chain reaction. If none are lost in non-fission reactions or by escaping to the outside world (prevention of which is the purpose of the tamper), then there will be N_0 “first generation” fissions, which will create $N_0 \nu$ neutrons. (Neutron loss can be modeled by decreasing the value of ν as desired.) These neutrons will lead to $N_0 \nu^2$ “second-generation” fissions and the creation of further $N_0 \nu^2$ neutrons and so on. The mathematics of such a progression shows that by the end of G such generations, the total number of fissions that will have occurred is given by

$$N_G = N_0 \left(\frac{\nu^G - 1}{\nu - 1} \right). \quad (9)$$

For any appreciable number of generations, the subtractive factor of 1 in the numerator of Eq. (9) can be discarded (see Exercise 4). If E_f is the energy liberated per fission (typically ~ 175 MeV), then the total energy liberated after G generations will be $E_f N_G$. Further, if τ is the average time for a single fission generation [Eq. (8)], then G generations will correspond to elapsed time $t = G\tau$. I now assume that all of the energy liberated until time t appears as the kinetic energy $M_{\text{tot}} v_{\text{core}}^2 / 2$ of the expanding (core + tamper) of total mass $M_{\text{tot}} = M_{\text{core}} + M_{\text{tamp}}$. Assuming that the (core + tamper) is expanding at a uniform rate throughout, the square of the expansion speed will be $(dr/dt)^2$, or

$$\left(\frac{dr}{dt} \right)^2 \sim \left[\frac{2 E_f N_0}{M_{\text{tot}} (\nu - 1)} \right] e^{(\ln \nu / \tau) t}, \quad (10)$$

where $\nu^G = \nu^{(t/\tau)}$ has been written as $e^{(\ln \nu / \tau) t}$. If, as above, R_{init} and R_{shut} are the initial (time zero) and second-criticality radii (time t_{shut}) of the core, then we can solve Eq. (10) for dr/dt and integrate to give

$$\int_{R_{\text{init}}}^{R_{\text{shut}}} dr \sim \sqrt{\frac{2E_f N_0}{M_{\text{tot}}(\nu - 1)}} \int_0^{t_{\text{shut}}} e^{(\ln \nu / 2\tau)t} dt. \quad (11)$$

Defining

$$\Delta R = R_{\text{shut}} - R_{\text{init}}, \quad (12)$$

and carrying out the integral then gives

$$t_{\text{shut}} \sim \left(\frac{2\tau}{\ln \nu} \right) \ln \left\{ (\Delta R) \left(\frac{\ln \nu}{2\tau} \right) \sqrt{\frac{M_{\text{tot}}(\nu - 1)}{2E_f N_0}} \right\}. \quad (13)$$

In deriving Eq. (13), the lower limit of integration of the time integral was dropped on the rationale that it will be small compared to the upper limit. Setting $G_{\text{shut}} = t_{\text{shut}}/\tau$ and computing the energy $E_f N_G$ liberated until shutdown with the help of Eq. (9) gives the yield as

$$E_{\text{shut}} \sim \left(\frac{E_f N_0}{\nu - 1} \right) \nu^{(t_{\text{shut}}/\tau)} \sim \frac{M_{\text{tot}}}{8} \left(\frac{\Delta R \ln \nu}{\tau} \right)^2. \quad (14)$$

This expression for E_{shut} arises from using Eq. (13) and can be derived most easily by considering the logarithm of E_{shut} . Interestingly, the initial number of neutrons N_0 cancels out in this calculation, as does E_f . If M_{tot} , ΔR , and τ are entered into Eq. (14) in kilograms, meters, and seconds, E_{shut} will emerge in Joules and can be converted to kilotons (kt) equivalent via the conversion factor $1 \text{ kt} = 4.2 \times 10^{12} \text{ J}$. Finally, if W represents the number of Joules of fission energy per kilogram of core material (which will implicitly contain a factor of E_f), the total energy potentially available from fissions will be $W M_{\text{core}}$, and the efficiency ε is defined as E_{shut} divided by $W M_{\text{core}}$

$$\varepsilon \sim \frac{M_{\text{tot}}}{8 W M_{\text{core}}} \left(\frac{\Delta R \ln \nu}{\tau} \right)^2. \quad (15)$$

These results give the same dependence of the efficiency on ΔR and τ as do more sophisticated analyses [see, for example, Eq. (70) of Ref. 5], but without having to invoke details of diffusion analyses. An important conceptual point here is that the yield depends on the inverse-square of the neutron travel time to fission. This is why nuclear weapons utilize *fast*-neutron chain reactions as opposed to thermal-neutron chain reactions: fission-generated neutrons travel at speeds $\sim 10^4$ that of thermal ones, so the yield of a slow-neutron bomb would, all other things being equal, be only $\sim 10^{-8}$ of that of a fast-neutron bomb. Making a slow-neutron bomb would be a waste of time, money, and material.

The question remains as to what value of τ to use in Eqs. (14) and (15). In evaluating the yield and efficiency, I use the value of τ at the time of second criticality. The rationale for this is that in a reaction where the number of neutrons doubles in each generation (which is approximately true for both uranium and plutonium cores), the number of neutrons created in a given generation is equal to the sum of the number of neutrons created in all preceding generations. Consequently, the vast majority of the energy liberation occurs in the last few generations, and it seems appropriate to evaluate the yield with the value of τ relevant to these generations.

Equation (14) provides a straightforward way to estimate the yield of any combination of core and tamper masses. To

begin, specify these masses and the nuclear properties of the core and tamper and set an initial value for the compression ratio. The initial radius R_{init} is calculated from Eq. (1). Trial-and-error values of the compression ratio at second criticality are tried in Eqs. (1)–(7) until Eq. (3) is satisfied, from which R_{shut} is determined via Eq. (1) using the final compression ratio; the yield and efficiency then follow. The final compression ratio will depend only on the core and tamper masses, not on the initial compression ratio.

IV. RESULTS

I have developed a one-page Excel spreadsheet to carry out the calculations described above (available as supplementary material⁶). Parameter values for the two practicable core materials, U-235 and Pu-239, are listed in Table I, and values for some typical tamper materials are listed in Table II. After entering core and tamper masses and parameters in units as indicated in the spreadsheet, along with an initial compression ratio, the user runs a single “Goal Seek” routine to determine the final compression ratio at second criticality and the estimated yield in kilotons and percent efficiency are automatically computed. The efficiency is computed on the basis of assuming $W = 17 \text{ kt}$ fission energy per kg of material, equivalent to an energy of 175 MeV per fission. The neutron speed is assumed to be that corresponding to kinetic energy 2 MeV , the average of fission-liberated neutrons; this can be changed by the user if desired. In what follows, this routine is applied to the *Little Boy* and *Fat Man* bombs of World War II. (Caution: If your value of C_{final} emerges as *greater* than C_{init} , your core/tamper configuration must have been subcritical and needs to be modified!)

A. Little Boy

The uranium-based *Little Boy* bomb dropped over Hiroshima comprised two pieces of U-235 that were assembled into a supercritical core by propelling a hollow cylindrical “projectile” piece initially located in the tail of the bomb forward to seat with a solid cylindrical “target” piece located in the nose; this is sketched in Fig. 5 of Ref. 5. The target was surrounded by a tungsten-carbide (WC) tamper, and the projectile was followed by a plug of the same material. The result of this was that when the core was fully assembled, it was essentially a solid cylinder of U-235 surrounded by a solid cylindrical tamper. This is sketched schematically in cross-section in Fig. 2, where the dimensions are adopted from John Coster-Mullen’s book on the construction and use of the bombs.⁸ The assembled core was 6.25 in. in diameter, but the tamper bore within which it resided was 6.5 in. in diameter, as indicated in the figure. With these dimensions, the assembled core had a volume of $214.76 \text{ in.}^3 = 3519.24 \text{ cm}^3$, and the tamper a volume of $2156.9 \text{ in.}^3 = 35345.3 \text{ cm}^3$. With respective densities of 18.71 and 15.63 g cm^{-3} , their masses were $M_{\text{core}} = 65.8 \text{ kg}$ and $M_{\text{tamp}} = 552.5 \text{ kg}$. (This tamper mass

Table I. Adopted parameters for U-235 and Pu-239. See Ref. 7, Sec. 2.2.

Quantity	Symbol	Unit	²³⁵ U	²³⁹ Pu
Atomic weight	A	g mol^{-1}	235.04	239.05
Density	ρ	g cm^{-3}	18.71	15.6
Fission cross-section	σ_{fiss}	bn	1.235	1.800
Elastic cross-section	σ_{elas}	bn	4.566	4.394
Neutrons per fission	ν	—	2.637	3.172

Table II. Adopted parameters for common tamper materials. See Sec. IV of Ref. 2 for source for cross-sections. Lead and tungsten both have several naturally occurring isotopes; values given here are abundance-weighted averages.

Tamper	Atomic weight (g mol ⁻¹)	Density (g cm ⁻³)	Elastic scattering cross-section (bn)
Aluminum	26.982	2.699	2.967
Beryllium oxide	25.01	3.02	5.412
Depleted U (U-238)	238.05	18.95	4.804
Lead	207.2	11.35	5.587
Tungsten carbide	195.85	15.63	6.587

differs from that used in Ref. 2 due to refined measurements published by Coster-Mullen subsequent to the publication of that paper.) This was a non-implosion bomb, so $C_{\text{init}} = 1$.

Running the spreadsheet shows that second criticality occurs at $C_{\text{final}} = 0.55$ after an expansion distance $\Delta R = 2.1$ cm and that the yield and efficiency are predicted to be 30.4 kt and 2.7%. Various estimates of the yield of *Little Boy* can be found, but a 1985 Los Alamos assessment gave 15 kt, about half of what is estimated here.⁹ This may seem a drastic discrepancy, but there are a number of reasons why the present model can be expected to overestimate the yield. First, I have assumed an idealized spherical geometry, which is likely to be somewhat more efficient than a cylindrical configuration. Second, the *Little Boy* core was likely not pure U-235. If we estimate $\sim 80\%$ U-235 purity and reduce the core mass to 52 kg U-235 (and ignore the “dead weight” of the remaining U-238, although there could well have been some U-238 fissions by high-energy neutrons), the yield prediction reduces to 21 kt. Third, there will surely be some loss of neutrons by inelastic scattering within the tamper material. On assuming a 52-kg core and reducing the number of neutrons per fission by about 10% to 2.4, the predicted yield drops to 12.0 kt. Predicted yields are very sensitive to input parameters: leaving the core mass at 65.8 kg but reducing ν to 2.25 is by itself sufficient to reduce the predicted yield to ~ 12.7 kt. A further effect not accounted for in the present model is depletion of the fissile material during the explosion, but for efficiencies of only a few percent, this will be a small issue.

B. Fat Man

In contrast to *Little Boy*, *Fat Man* was a spherical-geometry implosion weapon wherein a subcritical core of Pu-239 was squeezed to critical density by explosives configured to crush tamper shells that surrounded the core. A scale drawing of *Fat Man* can be found on p. 182 of

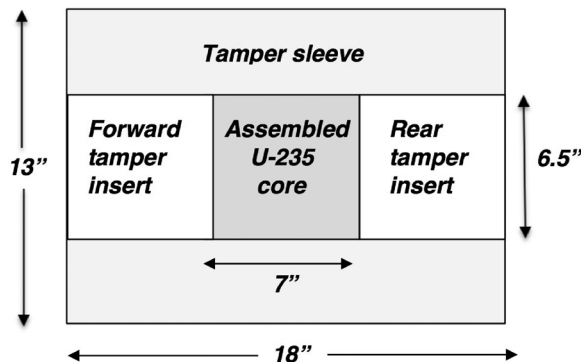


Fig. 2. Dimensions of cylindrical *Little Boy* core/tamper assembly as sketched in cross-section (dimensions from Ref. 7). Diagram is not to scale.

Coster-Mullen’s book. According to his dimensions, the core had a diameter of 3.62 in. or a volume of 407.0 cm³. However, from this must be subtracted the 4.4 cm³ volume of a 0.8-in. diameter polonium-beryllium neutron initiator housed at the center of the core; this leaves the core with a volume of 402.6 cm³ or a mass of 6.3 kg on assuming a density of 15.6 g cm⁻³ for Pu-239. The core was surrounded by a shell of depleted uranium (essentially pure U-238) of outer diameter 8.75 in., which was surrounded by an aluminum shell of inner diameter 9 in. and outside diameter 18.5 in. (the two shells were separated by a thin layer of plastic). With the densities given in Table II, these dimensions correspond to masses of 99.9 and 129.7 kg, respectively. Because the present model assumes only a single tamper shell, I ascribe the total tamper mass of 229.6 kg to a pure U-238 tamper.

The greatest unknown in modeling *Fat Man* is the initial compression ratio that was achieved by the implosion. A detailed analysis of the effects of the *Trinity* test indicates a value of ~ 2.5 , so I will work with numbers in this vicinity.¹⁰

With the above numbers and $C_{\text{init}} = 2.5$, the present model predicts a yield of 21.0 kt and an efficiency of 19.6%; the final compression ratio is 1.28. Remarkably, the Los Alamos report cited above quotes a yield for the Nagasaki *Fat Man* of 21 kt. This agreement must to some extent be fortuitous, as it has been estimated that some 20%–30% of the yield of this bomb (that is, ~ 4 –6 kt; Ref. 10) was contributed by high-energy neutrons inducing fissions in the U-238 tamper; perhaps this contribution offsets loss of neutrons by non-fission captures by U-238 and uncertainty in the compression ratio. Reducing C_{init} to 2.25 gives a yield of about 15.5 kt,

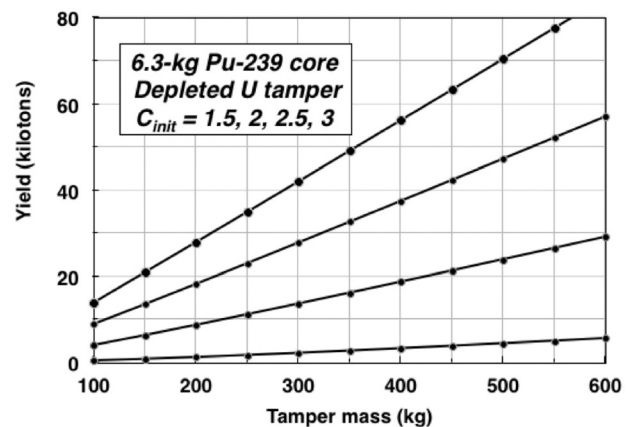


Fig. 3. Yield as calculated by the present model as a function of tamper mass for a *Fat Man*-type plutonium core with a U-238 tamper for initial compression ratios 1.5, 2, 2.5, and 3 (bottom to top curves).

roughly equivalent to the Pu-239-only yield if 75% of the overall yield was due to this isotope.

Figure 3 shows the results of further calculations for a *Fat Man* core, plotting estimated yield vs tamper mass for various initial compression ratios; the curves are interpolated. As indicated by Eq. (14), yield grows essentially linearly with tamper mass, with greater compressions giving greater growth rates.

V. SUMMARY

Despite the various approximations built into this model, it is encouraging to see that it gives results of about the correct magnitude for both Manhattan Project bombs. The predictions of this model obviously need to be taken with some caution, but it should be suitable as a pedagogical tool both for making numerical estimates and for reinforcing understanding of the parameters on which the yield depends. Nuclear weapons are not as mysterious as often thought; much of the physics underlying them is accessible to undergraduate students.

ACKNOWLEDGMENTS

I am grateful to Alex Wellerstein for drawing my attention to Ref. 10 and to the reviewers of this paper for a number of helpful suggestions.

APPENDIX: STUDENT EXERCISES

Exercise 1. Beryllium oxide (BeO) is a desirable tamper material because of its small neutron-capture cross-section. If a 50-kg U-235 core is surrounded with a 200-kg BeO tamper and imploded to $C_{\text{init}} = 1.5$, what is the estimated yield? [Ans: 25.8 kt]

Exercise 2. Terrorists have acquired 5 kg of Pu-239 and claim that they can make a crude implosion device with a 150-kg aluminum tamper and $C_{\text{init}} = 2$. What yield can be expected? [Ans: 2.8 kt]

Exercise 3. Suppose that some material has $\nu = 2.5$. What is the percentage change in the predicted yield if ν drops by 10% due to neutron loss, presuming that ΔR and τ are unaffected by the change in ν ? [Ans: About -22%]

Exercise 4. Suppose that a bomb yields 15 kt for $\nu = 2.5$ and $N_0 = 100$. If each fission liberates 175 MeV, how many fission generations occurred? [Ans: ~56–57]

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²C. Reed, “A Brief primer on tamped fission-bomb cores,” *Am. J. Phys.* **77**(8), 730–733 (2009).

³C. Reed, “Note on the minimum critical mass for a tamped fission bomb core,” *Am. J. Phys.* **83**(11), 969–971 (2015).

⁴C. Reed, “Resource Letter MP-3: The Manhattan Project and Related Nuclear Research,” *Am. J. Phys.* **84**(10), 734–745 (2016).

⁵C. Reed, “A physicist’s guide to *The Los Alamos Primer*,” *Phys. Scr.* **91**(11), 113002 (2016); “Erratum,” **91**(12), 129601 (2016).

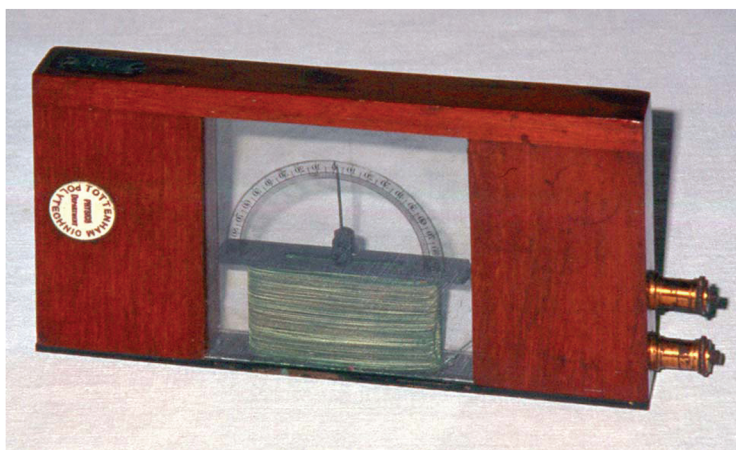
⁶See supplementary material at <http://dx.doi.org/10.1119/1.5009102> for user-supplied values in the spreadsheet appear in red. Note that masses must be entered in grams. Cell C43 is goal-seek adjusted to zero by adjusting on cell C36. (the final compression ratio).

⁷C. Reed, *The Physics of the Manhattan Project*, 3rd ed. (Springer, Heidelberg, 2015).

⁸J. Coster-Mullen, *Atom Bombs: The Top Secret Inside Story of Little Boy and Fat Man* (Coster-Mullen, Waukesha, WI, 2016).

⁹J. Malik, “The yields of the Hiroshima and Nagasaki nuclear explosions,” Los Alamos National Laboratory Report LA-8819, 1985. <<http://atomichive.com/Docs/pdfs/00313791.pdf>>.

¹⁰T. Semkow, P. Parekh, and D. Haines, “Modeling the effects of the Trinity test,” in *Applied Modeling and Computations in Nuclear Science*, American Chemical Society Symposium Series 945 (American Chemical Society, Washington, 2006), pp. 142–159.



Projection Galvanometer

The round label reads “Tottenham Polytechnic Physics Department.” This London institution was opened in 1897, and this was probably part of the original collection of apparatus. The projection meter is 3.5 in. in height to fit into the projection gate of British slide projectors. The galvanometer has to be used upright, for it of the Bréguet style in which a bar magnet, attached perpendicular to the needle, feels a torque when current passes through the coil. A counterweight below the magnet provides the gravitational torque that opposes the magnetic torque. See John A. Daffron and Thomas B. Greenslade, Jr., “A Modern Bréguet-Type Galvanometer”, *Phys. Teach.*, **53**, 289–289. The meter is in the collection of Jack Judson. (Picture and Notes by Thomas B. Greenslade, Jr., Kenyon College)